

Jaume Mora i Ladària

AI Engineer

Barcelona

☎ +34 699 25 09 31 • ✉ moraladariajaume@gmail.com • 🌐 jaumemil.cat
🐙 JaumeMiL • in jaumemil

Professional Summary

Artificial Intelligence graduate from the Universitat Politècnica de Catalunya with hands-on experience across **Natural Language Processing, Large Language Models, Computer Vision, Deep Learning and Biomedical AI**. Experienced in building and evaluating AI systems from data preparation and experimentation to model fine-tuning, quantitative evaluation, error analysis and deployment-oriented engineering.

Strong background in **Python, PyTorch, TensorFlow and OpenCV**, with practical experience in **Llama, Qwen, Hugging Face, LoRA/QLoRA, prompting, synthetic data generation, medical imaging, transfer learning and multimodal experimentation**.

Hands-on experience with remote Linux environments and high-performance computing infrastructure including **BOADA and MareNostrum**, using SSH, SLURM, shared-resource job scheduling and resource-aware execution. Particularly interested in NLP, computer vision and machine learning engineering roles in Barcelona.

Core Competencies

Computer Vision, Medical Imaging, Natural Language Processing, Large Language Models, LLM Fine-Tuning, LoRA/QLoRA, Prompt Engineering, Synthetic Data Generation, Deep Learning, Transfer Learning, Machine Learning Engineering, Model Evaluation, Error Analysis, Data Pipelines, High-Performance Computing, Testing and Technical LaTeX Documentation.

Professional Experience

SCOPIA - Hospital Universitari Son Espases & UIB

Palma

AI & Multimodal Machine Learning Research Assistant

06/2026 – 08/2026

- Designed and executed an end-to-end **PyTorch** experimental pipeline for wound-infection classification using a dataset of 394 medical images with strong class imbalance.
- Benchmarked multiple CNN and Transformer architectures, including **ResNet50, EfficientNet-B0, DenseNet121, InceptionV3, ViT-B/16, ConvNeXt-Tiny, Swin-T and Xception**, applying transfer learning, fine-tuning and threshold analysis.
- Improved infection classification through wound-region preprocessing and dilated wound masks; the best **ViT-B/16** configuration achieved **98.3% accuracy, 0.933 macro-F1 and 0.981 AUC** on the held-out test set.
- Investigated class imbalance, false-negative trade-offs, wound-context masking and model behaviour using quantitative metrics and failure-case analysis.
- Evaluated multimodal extensions incorporating clinical questionnaire variables, including direct fusion and post-hoc calibration strategies.
- Produced reproducible experimentation workflows, structured result comparisons and technical documentation for use within the REDSCAR research project.

Universitat Politècnica de Catalunya

Barcelona

Learning Grant Student — Python & Computer Vision

09/2025 – 06/2026

- Independently migrated approximately **30 computer vision and image-processing laboratories** from MATLAB to **Python** using NumPy, OpenCV and Matplotlib.
- Produced the teaching material in Catalan, Spanish and English, representing roughly **90 localized laboratory versions**.
- Refactored legacy procedural exercises into reusable Python workflows for image transformations, filtering, feature extraction, visualisation and quantitative analysis.
- Validated Python implementations against expected and legacy outputs, investigating numerical differences, regressions and edge cases.
- Produced technical documentation and supported adoption by students and teaching staff.

Outlier

Remote

Freelance Generative AI Evaluator

11/2024 – 01/2025

- Evaluated large language model outputs across factual accuracy, reasoning quality, relevance, completeness and instruction adherence.
- Performed hallucination detection, ambiguity analysis and structured comparison of alternative model responses.
- Identified recurring failure patterns including unsupported claims, contradictions, incomplete reasoning and unreliable assumptions.
- Produced structured written assessments explaining model limitations and evaluation criteria.

Canet Rock Festival

Bar Manager / Operational Lead

Canet de Mar

06/2022 – 07/2025

- Led staff coordination, logistics and operational workflows during large-scale festival events with more than **25,000 attendees**.

Education

Universitat Politècnica de Catalunya

BSc in Artificial Intelligence

Barcelona

2022 – 2026

Graduated: June 2026.

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Large Language Models, Generative AI, Statistics, Databases, Software Engineering, Object-Oriented Programming, Distributed Systems, Large-Scale Computing and MLOps.

Bachelor's Thesis

Universitat Politècnica de Catalunya

A Novel Computer Vision Framework to Enhance Robot Skills, **Grade: 9.5/10**

Barcelona

2026

- Designed and implemented an **end-to-end distributed computer vision system** using Python and ROS 2 for real-time gesture-based teleoperation of a TurtleBot3 Waffle Pi.
- Extracted 21 hand landmarks in real time using **MediaPipe Hands** and used them as input to a lightweight EurekaNet deep learning classifier trained on the IPN Hand dataset.
- Implemented the complete perception-to-control pipeline including image acquisition, preprocessing, neural network inference, ROS 2 communication and robot motion logic.
- Validated the system through **600 quantitative tests across 10 gesture classes** and qualitative testing with different users.
- Implemented safety and reliability mechanisms including confidence thresholds, temporal voting, gesture locking, emergency-stop gestures and LiDAR-based checks.
- Achieved real-time teleoperation with approximately **200–300 ms perceived latency over Wi-Fi**.
- Containerised the environment using **Docker and Rocker** to support reproducible execution across development systems.

Selected AI Projects & Academic Experience

Aspect-Based Sentiment Analysis with LLMs:

- Evaluated **Llama 3.2 3B and Qwen 2.5 3B** for aspect and sentiment extraction from restaurant reviews using zero-shot, few-shot, fine-tuning, synthetic data generation and ensemble strategies.
- Applied **LoRA/QLoRA and 4-bit quantization**, achieving a best individual result of **85.2 macro-F1**, compared with 63.4 in zero-shot and 76.4 in few-shot.

Catalan Dialect Robustness of Large Language Models:

- Designed an exploratory evaluation framework to assess how modern LLMs understand and generate Catalan dialectal variation.
- Created linguistic stress tests covering morphology, vocabulary, phraseology and informal regional language, and evaluated model outputs using a weighted rubric for grammaticality, dialect adequacy and task fidelity.

Opinion Detection in Film Reviews:

- Developed and compared supervised and unsupervised sentiment-analysis systems.
- Built a supervised pipeline using preprocessing, POS tagging, lemmatization, CountVectorizer and Random Forest, achieving **87.75% test accuracy and weighted F1**, and compared it against VADER and SentiWordNet-based approaches.

Language Identification with Naïve Bayes and Trigrams:

- Built a character-trigram language identification system using Naïve Bayes with Laplace smoothing for six European languages.

- Achieved approximately **99.9% global accuracy** and analysed confusion patterns between closely related language families.

End-to-End Data Pipeline for Barcelona Road Accidents:

- Built a complete DataOps pipeline using **Python, Apache Spark, DuckDB and Parquet**.
- Ingested Open Data BCN sources through the CKAN API and implemented Landing, Formatted, Trusted and Exploitation layers, followed by geospatial and risk-analysis pipelines.

Technical Skills

NLP & LLMs: **Llama, Qwen, Hugging Face, LoRA, QLoRA**, Prompt Engineering, Zero-shot and Few-shot Learning, Synthetic Data Generation, Sentiment Analysis, Aspect-Based Sentiment Analysis, Embeddings, Retrieval-Augmented Generation, Structured Outputs and LLM Evaluation

Machine Learning: **PyTorch, TensorFlow, scikit-learn**, Deep Learning, Transfer Learning, Fine-tuning, Feature Engineering, Model Validation, Generalisation Analysis and Error Analysis

Computer Vision: **OpenCV, MediaPipe**, Medical Imaging, Image Processing, Human Gesture Recognition, Feature Extraction, Wound Image Classification, Vision Transformers and Model Evaluation

Programming: **Python, C++, SQL, MATLAB**, Object-Oriented Programming, Git and GitHub

Data Engineering: Apache Spark, DuckDB, Parquet, Data Ingestion, ETL Workflows, Data Pipelines, Data Cleaning, Structured Data and Quantitative Analysis

HPC & Systems: **Linux, SSH, SLURM, BOADA, MareNostrum**, High-Performance Computing, Job Submission, Queue Management, Docker and Remote Development

Engineering: ROS 2, REST APIs, Distributed Systems, Testing, Debugging, Structured Logging, Reliability Engineering and Technical Documentation

Languages

Catalan: Native

Spanish: Native

English: Advanced professional proficiency

German: Basic

Awards & Additional Achievements

2024: **Second Place — Datathon 2024, UPC/FME**. Developed the Aider Assistant project in a competitive data and AI challenge, combining rapid prototyping, analytical reasoning, teamwork and technical communication.

Additional: Four-time Na Batlessa Catalan Literature Prize winner; competitive chess background; UPC Volleyball Team member; more than ten years of violin training.

Links

Portfolio: jaumemil.cat

GitHub: github.com/JaumeMiL

LinkedIn: linkedin.com/in/jaumemil